

Angular Developer → Full■Stack + AI + Cloud: 6■Month Upskilling Plan (India 2026)

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Summary

AI is reshaping (not replacing) software roles in India. Your competitive edge will come from combining modern Angular, full■stack backends (NestJS/Node), AI integration (LLMs, RAG), and cloud/DevOps with strong system design and observability. This PDF contains a 6■month plan, a flagship project brief, weekly routine, resume guidance, and interview prep checklist.

6■Month Roadmap

Month 1 — Angular + AI■Assisted Development

- Angular: Signals, standalone APIs, SSR with Angular Universal; strict typing; workspace organization.
- AI■assisted coding: Set up Copilot/Codeium; use prompts for refactors, tests, and docs; target 30–40% of routine code via AI.
- Deliverable: Refactor a feature using signals + SSR; unit tests generated/assisted by AI.

Month 2 — Backend (NestJS) to Become Full■Stack

- NestJS: modules/providers, configuration, validation (DTO/Zod), OpenAPI/Swagger.
- Data: Postgres with Prisma/TypeORM; Redis for cache/rate limiting; Auth with JWT/OAuth2.
- Deliverable: REST API + JWT auth + Postgres; integrate with Angular front■end.

Month 3 — Cloud & DevOps

- Choose Azure or AWS. Azure: App Service/Container Apps, Azure Database for PostgreSQL, Key Vault, App Insights. AWS: ECS/Fargate or Beanstalk, RDS Postgres, Secrets Manager, CloudWatch.
- CI/CD: GitHub Actions pipelines (lint, test, build, deploy); basic Terraform for infra as code.
- Deliverable: One■click deploy from main → production with monitoring.

Month 4 — Performance & Accessibility

- Performance: Route■level code splitting, lazy loading, prefetch, image optimization, Core Web Vitals.
- Accessibility: ARIA roles, keyboard flows; automated checks (axe).
- Deliverable: Lighthouse ≥ 90 on Performance and Accessibility across key pages.

Month 5 — AI Integration in the Product

- LLM features: Server■side LLM proxy in NestJS (OpenAI/Anthropic/Mistral).
- RAG basics: Vector store (pgvector/Weaviate) for domain docs; semantic search.
- UX: Streaming responses (SSE), guardrails, prompt templates, telemetry.

- Deliverable: Production-ready AI feature: semantic search or copilot assistant with citations.

Month 6 — Architecture, Observability, Interview Prep

- Architecture: Caching, queues, idempotency, rate limiting, circuit breakers.
- Observability: Structured logs, metrics, traces; dashboards and alerts; SLOs and error budgets.
- Interview: System design drills; architecture document; on-call scenarios.
- Deliverable: Architecture README + diagrams + runbooks; portfolio case study.

Flagship Portfolio Project — ‘Insight Hub’

A secure knowledge portal where teams upload documents and query them via semantic search and a chat assistant that cites sources. This demonstrates full-stack engineering, AI integration (RAG), performance, security, and cloud deployment in a single, cohesive product.

Scope & Modules

- Auth & Roles: Email/password + OAuth; admin/user roles.
- Ingestion: Upload PDFs/Markdown; server-side text extraction.
- Indexing (RAG): Chunking to Postgres + pgvector; access-control metadata.
- Search: Hybrid (keyword + vector) with facets and filters.
- Chat Assistant: Angular chat UI → NestJS → LLM provider; SSE streaming; token/latency metrics.
- Observability: App Insights/CloudWatch; trace IDs FE → API → LLM calls.
- Security: Rate limits, validation, output filtering; secrets in Key Vault/Secrets Manager.
- Performance: SSR, route prefetching, code-split routes; Lighthouse goals ≥ 90 .

Milestones (Map to Months)

- M1–M2: Angular shell + NestJS API + Postgres; auth + core CRUD.
- M3: CI/CD + cloud infra + basic observability.
- M4: Performance & accessibility; CWV targets met.
- M5: LLM chat + RAG search + streaming UI.
- M6: System design doc, dashboards, runbook, final demo video.

Weekly Routine (6–8 hrs/week)

- Mon–Tue (2h): Concepts + short notes.
- Wed (1–2h): Implement one focused feature (use AI assistant).
- Thu (1h): Tests + refactor (ask AI for alternatives).
- Sat (2–3h): Integrate, document, and deploy.
- Biweekly: Record a 3–5 min demo video for your portfolio.

Resume Refresh (AI Era)

- Summary: Sr. Software Engineer (Angular/Full Stack) building AI-enabled products on Azure; led SSR migrations, improved CWV, shipped LLM features with CI/CD and observability.
- Experience bullets (add metrics): LCP 4.2s → 1.9s; p95 API 220ms; uptime 99.9%; tickets –18%; PR→prod lead time 2.3h; MTTR 90m → 22m.
- Skills: Angular/Universal/RxJS/NgRx • Node/NestJS/REST/GraphQL/Postgres/Redis • LLM APIs/RAG • Azure/AWS/Docker/Terraform/GitHub Actions/Observability.

Interview Readiness (Checklist)

- Architecture narrative for 'Insight Hub' (10–12 min).
- Tradeoffs: SSR vs CSR; Redis vs DB cache; RAG vs keyword; cost vs latency.
- AI guardrails: PII handling, rate limits, abuse prevention, evals.
- System design drills: chat with streaming; multi-tenant document search with ACLs.

Notes & Market Context (India 2025–2026)

- AI/ML roles rose sharply in India while non-AI tech roles still grew; software engineering remains robust but more selective.
- India's demand for AI-skilled professionals is surging toward 2026; blending AI with full-stack skills maximizes opportunity.
- Software engineering job market trends: strong growth, AI tools used by most developers, portfolios and system design matter.